

Database Design



Status

Done

Conceptual Model

List of Entities

Entity Name	Description
Users	Data pengguna sistem (Admin Website, Admin Pusat, Admin Pos)
Roles	Role/level akses user
POS	Cabang bank sampah
Nasabah	Data nasabah per POS
Vendor	Mitra pengepul/vendor
POS_Vendor	Relasi POS dan Vendor (M:N)
WasteCategory	Kategori sampah (±5 kategori)
WasteType	Jenis sampah (±44 item)
VendorPrice	Harga sampah per vendor
PriceHistory	Riwayat perubahan harga
TransaksiPenimbangan	Transaksi setoran sampah
DetailPenimbangan	Detail item sampah dalam transaksi
TransaksiPenarikan	Transaksi penarikan saldo
AuditLog	Log aktivitas sistem
ImportLog	Log proses upload Excel

Relationships

Entity 1	Entity 2	Cardinality	Description
Roles	Users	1 : M	Satu role dimiliki banyak user
POS	Users	1 : M	Satu POS memiliki banyak Admin Pos
POS	Nasabah	1 : M	Satu POS memiliki banyak nasabah
Vendor	POS_Vendor	1 : M	Vendor bisa ke banyak POS
POS	POS_Vendor	1 : M	POS punya banyak vendor
Vendor	VendorPrice	1 : M	Vendor memiliki banyak harga
WasteType	VendorPrice	1 : M	Setiap jenis punya harga per vendor
VendorPrice	PriceHistory	1 : M	Harga punya histori
WasteCategory	WasteType	1 : M	Kategori memiliki banyak jenis
Nasabah	TransaksiPenimbangan	1 : M	Nasabah setor sampah
Nasabah	TransaksiPenarikan	1 : M	Nasabah tarik saldo
TransaksiPenimbangan	DetailPenimbangan	1 : M	Satu transaksi banyak detail
WasteType	DetailPenimbangan	1 : M	Jenis sampah digunakan di transaksi
Users	TransaksiPenimbangan	1 : M	Dicatat oleh admin
Users	TransaksiPenarikan	1 : M	Dicatat oleh admin
Users	AuditLog	1 : M	Aktivitas user tercatat
Users	ImportLog	1 : M	Upload dilakukan oleh user

List of Attributes

Entity: Users

Field	Type	Constraints	Description
user_id	VARCHAR(20)	PK	Custom ID (e.g., WW-AP-001)
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password_hash	TEXT	NOT NULL	Hashed password
role	VARCHAR(20)	NOT NULL	(ADMIN_WEB, ADMIN_PUSAT, ADMIN_POS)
pos_id	VARCHAR(20)	FK → pos.pos_id (nullable)	Only for Admin Pos
is_active	BOOLEAN	DEFAULT TRUE	Status akun
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp

Entity: POS

Field	Type	Constraints	Description
pos_id	VARCHAR(20)	PK	ID Pos
pos_name	VARCHAR(100)	NOT NULL	Nama pos
address	TEXT	NOT NULL	Alamat
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp

Entity: Vendor

Field	Type	Constraints	Description
vendor_id	VARCHAR(20)	PK	ID Vendor
vendor_name	VARCHAR(100)	NOT NULL	Nama vendor

contact	VARCHAR(100)		Kontak
created_at	TIMESTAMP	DEFAULT NOW()	Timestamp

Entity: Pos_Vendor

Field	Type	Constraints	Description
pos_id	VARCHAR(20)	PK, FK → pos.pos_id	
vendor_id	VARCHAR(20)	PK, FK → vendors.vendor_id	

Entity: Waste_Categories

Field	Type	Constraints	Description
category_id	VARCHAR(10)	PK	Category ID
category_name	VARCHAR(50)	NOT NULL	(Plastic, Paper, etc.)

Entity: Waste_Types

Field	Type	Constraints	Description
waste_type_id	VARCHAR(20)	PK	ID jenis sampah
category_id	VARCHAR(10)	FK → waste_categories.category_id	
waste_name	VARCHAR(100)	NOT NULL	Nama sampah

Entity: Vendor_Prices

Field	Type	Constraints	Description
price_id	SERIAL	PK	
vendor_id	VARCHAR(20)	FK → vendors.vendor_id	
waste_type_id	VARCHAR(20)	FK → waste_types.waste_type_id	
price	NUMERIC(12,2)	NOT NULL	Harga per kg
updated_at	TIMESTAMP	DEFAULT NOW()	

Entity: Nasabah

Field	Type	Constraints	Description
customer_id	VARCHAR(20)	PK	Nomor induk nasabah
pos_id	VARCHAR(20)	FK → pos.pos_id	
name	VARCHAR(100)	NOT NULL	Nama
address	TEXT	NOT NULL	
phone	VARCHAR(20)		
balance	NUMERIC(12,2)	DEFAULT 0	Saldo
is_active	BOOLEAN	DEFAULT TRUE	Soft delete
created_at	TIMESTAMP	DEFAULT NOW()	

Entity: Transactions

Field	Type	Constraints	Description
transaction_id	SERIAL	PK	
customer_id	VARCHAR(20)	FK → customers.customer_id	
pos_id	VARCHAR(20)	FK → pos.pos_id	
vendor_id	VARCHAR(20)	FK → vendors.vendor_id	
transaction_date	TIMESTAMP	DEFAULT NOW()	
total_amount	NUMERIC(12,2)	NOT NULL	Total uang
created_by	VARCHAR(20)	FK → users.user_id	

Entity: Transaction_Details

Field	Type	Constraints	Description
detail_id	SERIAL	PK	
transaction_id	INT	FK → transactions.transaction_id	

waste_type_id	VARCHAR(20)	FK → waste_types.waste_type_id	
weight	NUMERIC(10,2)	NOT NULL	Berat
price	NUMERIC(12,2)	NOT NULL	Harga saat transaksi
subtotal	NUMERIC(12,2)	NOT NULL	

Entity: Withdrawals

Field	Type	Constraints	Description
withdrawal_id	SERIAL	PK	
customer_id	VARCHAR(20)	FK → customers.customer_id	
amount	NUMERIC(12,2)	NOT NULL	
withdrawal_date	TIMESTAMP	DEFAULT NOW()	
created_by	VARCHAR(20)	FK → users.user_id	

Entity: Audit_Logs

Field	Type	Constraints	Description
log_id	SERIAL	PK	
user_id	VARCHAR(20)	FK → users.user_id	
action	VARCHAR(100)	NOT NULL	
entity	VARCHAR(50)	NOT NULL	
entity_id	VARCHAR(50)		
timestamp	TIMESTAMP	DEFAULT NOW()	

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